DRL Sessions 4 & 5 — MDP Formulation & Dynamic Programming (Exam Handout)

Session 4: MDP, 3 signals, returns, discounting, G_t recursion — Session 5: Policy Evaluation, Improvement, Iteration, Value Iteration, GPI, Async DP

Session 4 — MDP Formulation

1 · What is an MDP?

Markov Decision Process (MDP)

An MDP is a mathematically precise formulation of the RL problem. It consists of:

$$\langle \mathcal{S},\ \mathcal{A},\ \mathcal{R},\ p,\ \gamma \rangle$$

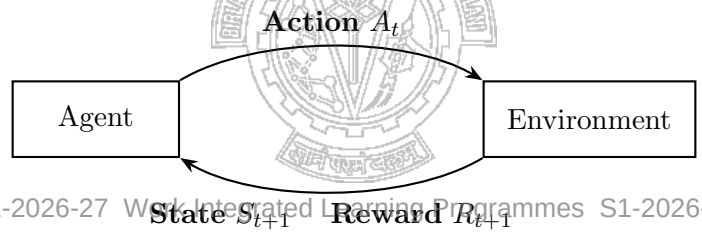
- $\mathcal{S}$ — finite set of **states**
- $\mathcal{A}$ — finite set of **actions**
- $\mathcal{R}$ — set of **rewards** (real-valued scalars)
- $p(s', r \mid s, a)$ — **dynamics function** (transition + reward probabilities)
- γ — **discount factor**, $0 \leq \gamma \leq 1$

The Markov Property: The future depends *only* on the current state, not on the history.

$$P(S_{t+1}, R_{t+1} \mid S_t, A_t) = P(S_{t+1}, R_{t+1} \mid S_0, A_0, \ldots, S_t, A_t)$$

2 · The 3 Signals Between Agent and Environment

Three Signals — Must Know



Signal	Direction	Meaning
S_t (State)	Environment → Agent	Current situation the agent observes
A_t (Action)	Agent → Environment	Decision taken at step t
R_{t+1} (Reward)	Environment → Agent	Scalar feedback after taking A_t in S_t

Dynamics function: $p(s', r \mid s, a) = \Pr\{S_{t+1} = s',\ R_{t+1} = r \mid S_t = s,\ A_t = a\}$

3 · Returns, Discounting, and G_t

Goal: Maximise the **expected return** G_t — not just the next reward, but the cumulative future reward.

Discounted Return

$$G_t \ = \ R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots \ = \ \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Role of γ (discount factor):

γ	Effect	Use case
$\gamma = 0$	Only immediate reward matters (myopic)	Short-horizon tasks; immediate payoff
$0 < \gamma < 1$	Future rewards discounted exponentially	Most practical RL tasks
$\gamma = 1$	All future rewards count equally	Episodic tasks with finite horizon only

Why discount?

1. Mathematical convergence: guarantees G_t is finite when rewards are bounded.
2. Economic interpretation: a reward now is worth more than the same reward later (time preference).
3. Uncertainty: future transitions may not occur as expected; discounting hedges against this.

4 · Episodic vs. Continuing Tasks

Two Task Types

	Episodic	**Continuing**
Termination	Has a terminal state T	Runs forever
Return formula	$G_t = \sum_{k=0}^{T-t-1} R_{t+k+1}$ (no γ needed)	$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$ (need $\gamma < 1$)
Examples	Chess, Tic-Tac-Toe, maze	Trading bot, robot control
$\gamma = 1$ allowed?	Yes (finite sum)	No (infinite sum diverges)

Unified formula (using absorbing terminal state with zero reward):

$$G_t = \sum_{k=t+1}^{T} \gamma^{k-t-1} R_k \quad \text{(works for both; set } T = \infty \text{ for continuing)}$$

5 · Recursive Formulation — G_t in terms of G_{t+1}

Key Recursion — used in every RL algorithm

$$G_t \;=\; R_{t+1} + \gamma G_{t+1}$$

Derivation (one step):

$$\begin{aligned} G_t &= R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots \\ &= R_{t+1} + \gamma\big(R_{t+2} + \gamma R_{t+3} + \cdots\big) \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

Why is this useful?

- Allows computing G_t *backwards* from the terminal state — no need to store the full trajectory.
- Directly motivates the TD update: estimate $G_t \approx R_{t+1} + \gamma V(S_{t+1})$.
- The Bellman equations (next section) are built entirely on this recursion.

6 · Value Functions and Bellman Equation

State-value and action-value functions

$$v_\pi(s) \;=\; \mathbb{E}_\pi[G_t \mid S_t = s] \;=\; \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a)\big[r + \gamma v_\pi(s')\big]$$

$$q_\pi(s, a) \;=\; \mathbb{E}_\pi[G_t \mid S_t = s,\, A_t = a] \;=\; \sum_{s',r} p(s', r|s, a)\big[r + \gamma v_\pi(s')\big]$$

The first equation is the **Bellman equation** for v_π.

Session 5 — Dynamic Programming (DP)

7 · Characteristics of DP

What is DP in the RL context?

DP refers to a collection of algorithms that use the **Bellman equations** to compute optimal value functions, given a **perfect model** of the environment.

Characteristics:

1. Requires a **complete model**: $p(s', r \mid s, a)$ must be fully known.
2. Computationally **expensive** for large state spaces, but **exact**.
3. Uses **bootstrapping**: updates are based on estimates of other states (not sampled rewards alone).
4. Finds the **optimal policy** π^* and optimal value function v^*.

When DP is applicable:

- State and action spaces are **finite** and **small**.
- The full transition model $p(s', r|s, a)$ is **known** (not needed to interact with real world).
- Examples: grid worlds, small board games, Jack's Car Rental, Gambler's Problem.
- Not applicable: large/continuous spaces (use function approximation), unknown model (use MC or TD).

8 · Key Terms (Policy Evaluation → Value Iteration)

Five Key Terms — Know Each Precisely

(1) Policy Evaluation (Prediction Problem)
Given policy π, compute $v_\pi(s)$ for all states. Use iterative sweeps:

$$v_{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a)\big[r + \gamma v_k(s')\big]$$

Repeat until $\max_s |v_{k+1}(s) - v_k(s)| < \theta$ (some small threshold).

(2) Policy Improvement (Control — one step)
Given v_π, form a *greedy* policy π':

$$\pi'(s) = \arg\max_a \sum_{s',r} p(s', r|s, a)\big[r + \gamma v_\pi(s')\big]$$

Policy Improvement Theorem: $v_{\pi'}(s) \geq v_\pi(s)$ for all s (the new greedy policy is at least as good as the old one).

(3) Policy Iteration
Alternate: full Policy Evaluation → full Policy Improvement → repeat.
Converges to π^* in a finite number of iterations (finite MDP).

(4) Value Iteration
Combine one sweep of (partial) evaluation + improvement:

$$v_{k+1}(s) = \max_a \sum_{s',r} p(s', r|s, a)\big[r + \gamma v_k(s')\big]$$

No need to wait for full policy evaluation to converge between improvements. Faster in practice.

(5) Generalized Policy Iteration (GPI)
The general idea of *any* interleaving of evaluation and improvement, regardless of granularity. All RL algorithms are a form of GPI.

Asynchronous DP

Standard DP sweeps all states uniformly (synchronous). **Asynchronous DP** updates states in any order — it can focus on states that are most relevant (e.g., those the agent visits often).

- Useful for large state spaces.
- Still requires a full model.
- Converges as long as every state is updated infinitely often in the limit.

9 · Policy Iteration vs. Value Iteration

	Policy Iteration	**Value Iteration**
Evaluation step	Run full Bellman expectation until convergence	One Bellman *optimality* sweep
Improvement step	Full greedy improvement after evaluation	Combined into same sweep
Iterations	Fewer policy updates, each costly	More iterations but each cheap
Update per step	$v_{k+1}(s) = \sum_a \pi(a\|s)[\ldots]$	$v_{k+1}(s) = \max_a[\ldots]$
Convergence	Finite (finite MDP)	Geometric (γ^k contraction)
When to prefer	When evaluation is cheap; small MDP	When model is known but large

10 · Worked Example — Value Iteration (Race Car / 1-D Track)

Race Car Problem (Simplified)

Setup: A car is on a 1-D track: states $s \in \{0, 1, 2, 3, 4\}$ (positions). State 4 = **Finish** (terminal). Actions: `Fast` (move 2 positions forward, reward -1) or `Slow` (move 1 position forward, reward -1). Goal: reach state 4 in minimum steps. Discount $\gamma = 0.9$. All transitions are deterministic.

Initialise: $v_0(s) = 0$ for all s. $v(4) = 0$ always (terminal).

Iteration 1 — Value Iteration sweep (apply $v_{k+1}(s) = \max_a[r + \gamma v_k(s')]$):
From state 3: Fast → 5 (out of range/clamped to 4), Slow → 4. Both reach finish: $r = -1$, $v_0(4) = 0$.

$$v_1(3) = \max(-1 + 0.9 \times 0,\ -1 + 0.9 \times 0) = -1$$

From state 2: Fast → 4 ($r = -1$, $v_0(4) = 0$); Slow → 3 ($r = -1$, $v_0(3) = 0$).

$$v_1(2) = \max(-1 + 0,\ -1 + 0) = -1$$

From state 1: Fast → 3 ($r = -1$, $v_0(3) = 0$); Slow → 2 ($r = -1$, $v_0(2) = 0$).

$$v_1(1) = -1$$

From state 0: Fast → 2; Slow → 1. Both: -1.

$$v_1(0) = -1$$

Iteration 2:
From state 3: same as before $= -1$.
From state 2: Fast → 4 ($-1 + 0 = -1$); Slow → 3 ($-1 + 0.9 \times (-1) = -1.9$).

$$v_2(2) = \max(-1,\ -1.9) = -1 \quad \Rightarrow \text{prefer } \texttt{Fast}$$

From state 1: Fast → 3 ($-1 + 0.9 \times (-1) = -1.9$); Slow → 2 ($-1 + 0.9 \times (-1) = -1.9$).

$$v_2(1) = -1.9$$

From state 0: Fast → 2 ($-1 + 0.9 \times (-1) = -1.9$); Slow → 1 ($-1 + 0.9 \times (-1.9) = -2.71$).

$$v_2(0) = \max(-1.9,\ -2.71) = -1.9 \quad \Rightarrow \text{prefer } \texttt{Fast}$$

Emerging policy: always take `Fast` — reach finish in fewest steps. Values converge downward as they properly account for the number of steps to terminal.

11 · Generalized Policy Iteration (GPI) — The Big Picture

GPI — All RL is GPI

$$\pi \xrightarrow{\text{Evaluate}} v_\pi \xrightarrow{\text{Improve}} \pi' \xrightarrow{\text{Evaluate}} v_{\pi'} \xrightarrow{\text{Improve}} \cdots \rightarrow \pi^*, v^*$$

- **Evaluation** drives v toward v_π (makes value consistent with current policy).
- **Improvement** drives π toward greedy w.r.t. current v (makes policy consistent with value).
- These two goals are *competing* (each step invalidates the other) but converge to a stable solution where $\pi = \text{greedy}(v_\pi) = \pi^*$.
- Every RL method — MC, TD, actor-critic — is a specific instance of GPI.

12 · Exam Key-Takeaways

Must Know — Sessions 4 & 5

1. **3 signals:** state S_t (env→agent), action A_t (agent→env), reward R_{t+1} (env→agent).
2. **Discount γ:** $\gamma = 0$ myopic; $\gamma = 1$ only for episodic; $0 < \gamma < 1$ in practice.
3. **Episodic vs continuing:** episodic has terminal state T; continuing needs $\gamma < 1$ for convergence.
4. **G_t recursion:** $G_t = R_{t+1} + \gamma G_{t+1}$ — memorise and be able to derive.
5. **Bellman equation:** $v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a)[r + \gamma v_\pi(s')]$.
6. **Policy Eval:** use Bellman expectation iteratively for given π.
7. **Value Iter:** use Bellman *optimality* — $\max_a$ replaces $\sum_a \pi(a|s)$.
8. **Policy Iter vs Value Iter:** Policy Iter waits for full eval; Value Iter doesn't.
9. **GPI:** any interleaving of evaluation + improvement converges to π^*.
10. **Async DP:** update states in any order; still needs full model; useful for large $|\mathcal{S}|$.

DRL EC3 Exam Handout — Sessions 4 & 5 (MDP & Dynamic Programming). Ref: Sutton & Barto Ch. 3–4.